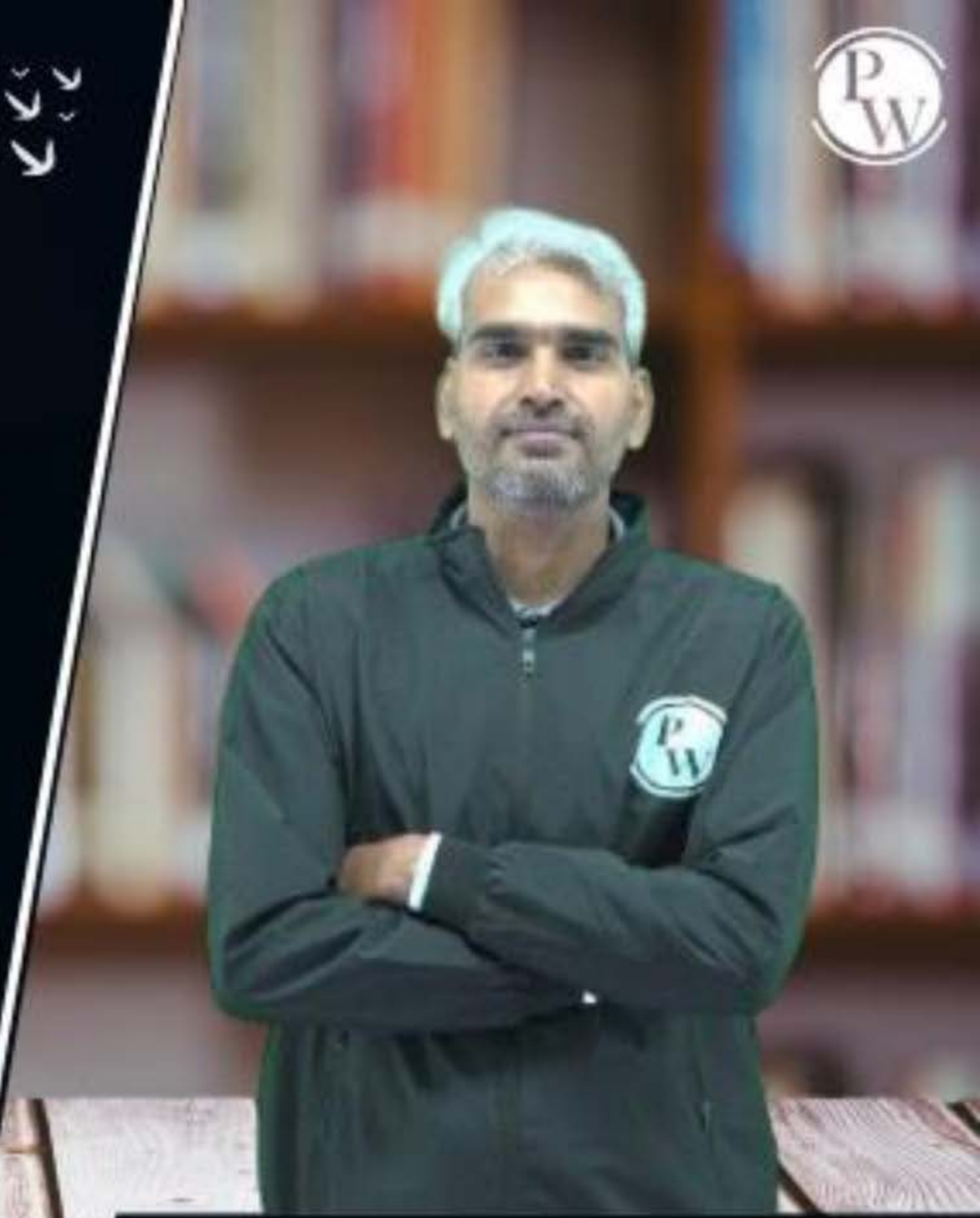
# Computer Science & IT

## Discrete Mathematics

**Set Theory & Algebra**

**Lecture No. 14**



**By- Vishal Sir**

# Recap of Previous Lecture

**Topic** Practice questions & PYQ

Slide

# Topics to be Covered

- **Topic** Basic concepts associated with group
- **Topic** Finite groups
- **Topic** Addition modulo m
- **Topic** Multiplication modulo m

Slide

# Topic : Special Sets

- N = Set of all natural numbers
- Z = Set of all integers
- R = Set of all real numbers
- Q = Set of all rational numbers
- $Q^*$ = Set of all non-zero rational numbers

Slide

# Groupoid

- Let 'S' be any non-empty set, and '$*$' is some binary operation, then set S w.r.t. binary opⁿ '$*$' is Called a groupoid if and only if

$$\boxed{a * b \in S, \ \forall a, b \in S}$$

↳ it is called Closure property.

i.e. Set S is Closed w.r.t. binary opⁿ '$*$'

Set 'S' w.r.t. binary opⁿ '$*$' Can be denoted by $(S, *)$

| | Algebraic Structure (Groupoid) | Semi-Group | Monoid | Group | Abelian Group |
|---|---|---|---|---|---|
| $\underline{N}$ $(N,+)$ | ✓ | | | | |
| $(N,\cdot)$ | ✓ | | | | |
| $(N,-)$ | ✗ | $(2-5)=-3 \notin N$ | | | |
| $(N,\div)$ | ✗ | | | | |
| $\underline{Z}$ $(Z,+)$ | ✓ | | | | |
| $(Z,\cdot)$ | ✓ | | | | |
| $(Z,-)$ | ✓ | | | | |
| $(Z,\div)$ | ✗ | $2\div 3 \notin Z$ | | | |
| $\underline{Q}$ $(Q,+)$ | ✓ | | | | |
| $(Q,\cdot)$ | ✓ | | | | |
| $(Q,-)$ | ✓ | | | | |
| $(Q,\div)$ | ✗ $0\in Q$, & $0\div 0$ not defined | | | | |
| $\underline{Q^*}$ $(Q^*,+)$ | ✗ $\frac{p}{q}+\left(\frac{-p}{q}\right)=0\notin Q^*$ | | | | |
| $(Q^*,\cdot)$ | ✓ | | | | |
| $(Q^*,-)$ | ✗ $\frac{p}{q}-\left(\frac{p}{q}\right)=0\notin Q^*$ | | | | |
| $(Q^*,\div)$ | ✓ | | | | |

Slide

## Topic : Semi-group

A groupoid $(S, *)$ is called a semi-group if and only if following property holds true

$$\boxed{(a*b)*c = a*(b*c)\ ,\ \forall a, b, c \in S}$$

↳ i.e. operation must be associative

Slide

$\mathbb{N} = \{1, 2, \ldots\}$

$\mathbb{Z}$

$\mathbb{Q}$

$\mathbb{Q}^*$

| | Algebraic Structure (Groupoid) | Semi-Group | Monoid | Group | Abelian Group |
|---|---|---|---|---|---|
| (N, +) | ✓ | ✓ | | | |
| (N, .) | ✓ | ✓ | | | |
| (N, −) | ✗ | ✗ | | | |
| (N, ÷) | ✗ | ✗ | | | |
| (Z, +) | ✓ | ✓ | | | |
| (Z, .) | ✓ | ✓ | | | |
| (Z, −) | ✓ | ✗ | | | |
| (Z, ÷) | ✗ | ✗ | | | |
| (Q, +) | ✓ | ✓ | | | |
| (Q, .) | ✓ | ✓ | | | |
| (Q, −) | ✓ | ✗ | | | |
| (Q, ÷) | ✗ | ✗ | | | |
| (Q*, +) | ✗ | ✗ | | | |
| (Q*, .) | ✓ | ✓ | | | |
| (Q*, −) | ✗ | ✗ | | | |
| (Q*, ÷) | ✓ | ✗ | | | |

Slide

# Topic : Monoid

A semi-group $(S, *)$ is called a Monoid if and only if there exists an element "$e \in S$"

s.t

$$\boxed{\begin{array}{l} a * e = a, \ \forall a \in S \\ \text{and} \\ e * a = a, \ \forall a \in S \end{array}}$$

↳ Such an element 'e' is called identity element w.r.t. binary opⁿ '*'

Slide

$N = \{1, 2, \ldots\}$

$Z$

$Q$

$Q^*$

| | Algebraic Structure (groupoid) | Semi-Group | Monoid | Group | Abelian Group |
|---|---|---|---|---|---|
| $(N, +)$ | ✓ | ✓ | $0 \notin N$ ✗ | | |
| $(N, \cdot)$ | ✓ | ✓ | $1 \in N$ ✓ | | |
| $(N, -)$ | ✗ | ✗ | ✗ | | |
| $(N, \div)$ | ✗ | ✗ | ✗ | | |
| $(Z, +)$ | ✓ | ✓ | $0 \in Z$, ✓ | | |
| $(Z, \cdot)$ | ✓ | ✓ | $1 \in Z$, ✓ | | |
| $(Z, -)$ | ✓ | ✗ | ✗ | | |
| $(Z, \div)$ | ✗ | ✗ | ✗ | | |
| $(Q, +)$ | ✓ | ✓ | $0 \in Q$, ✓ | | |
| $(Q, \cdot)$ | ✓ | ✓ | $1 \in Q$, ✓ | | |
| $(Q, -)$ | ✓ | ✗ | ✗ | | |
| $(Q, \div)$ | ✗ | ✗ | ✗ | | |
| $(Q^*, +)$ | ✗ | ✗ | ✗ | | |
| $(Q^*, \cdot)$ | ✓ | ✓ | $1 \in Q^*$, ✓ | | |
| $(Q^*, -)$ | ✗ | ✗ | ✗ | | |
| $(Q^*, \div)$ | ✓ | ✗ | ✗ | | |

Slide

## Topic : Group

A monoid $(S, *)$ is called a group, if and only if. for every element $a \in S$ there exists some element $b \in S$, such that

$$\boxed{\begin{array}{c} a * b = e \text{ (identity)} \\ \text{(and)} \\ b * a = e \end{array}}$$

↳ Here 'a' & 'b' are called inverse of each other

i.e. for a group, inverse of every element of the set must exist within the set itself!

Slide

| | | Algebraic Structure (Groupoid) | Semi-Group | Monoid | Group | Abelian Group |
|---|---|---|---|---|---|---|
| $\mathbb{N}$ | (N, +) | ✓ | ✓ | $0 \notin \mathbb{N}$ ✗ | ✗ | |
| | (N, ·) | ✓ | ✓ | $1 \in \mathbb{N}$ ✓ | ✗ inverse does not exist for any element Except for '1' | |
| | (N, −) | ✗ | ✗ | ✗ | ✗ | |
| | (N, ÷) | ✗ | ✗ | ✗ | ✗ | |
| $\mathbb{Z}$ | (Z, +) | ✓ | ✓ | $0 \in \mathbb{Z}$, ✓ | ✓ | |
| | (Z, ·) | ✓ | ✓ | $1 \in \mathbb{Z}$, ✓ | ✗ inverse does not exist for any element Except for '1' & '−1' | |
| | (Z, −) | ✓ | ✗ | ✗ | ✗ | |
| | (Z, ÷) | ✗ | ✗ | ✗ | ✗ | |
| $\mathbb{Q}$ | (Q, +) | ✓ | ✓ | $0 \in \mathbb{Q}$, ✓ | ✓ | |
| | (Q, ·) | ✓ | ✓ | $1 \in \mathbb{Q}$, ✓ | ✗, $0 \in \mathbb{Q}$, A set containing '0' can never be a group w.r.t. multiplication. | |
| | (Q, −) | ✓ | ✗ | ✗ | ✗ | |
| | (Q, ÷) | ✗ | ✗ | ✗ | ✗ | |
| $\mathbb{Q}^*$ | (Q*, +) | ✗ | ✗ | ✗ | ✗ | |
| | (Q*, ·) | ✓ | ✓ | $1 \in \mathbb{Q}^*$, ✓ | ✓ | |
| | (Q*, −) | ✗ | ✗ | ✗ | ✗ | |
| | (Q*, ÷) | ✓ | ✗ | ✗ | ✗ | |

Slide

A group $(S, *)$ is called an abelian group if and only if,

$$\boxed{a * b = b * a \ , \ \forall a, b \in S}$$

↳ i.e, $op^n$ '$*$' must be commutative on the elements of set S.

Slide

| | | Algebraic Structure (Groupoid) | Semi-Group | Monoid | Group | Abelian Group |
|---|---|---|---|---|---|---|
| $\mathbb{N}$ | $(N, +)$ | ✓ | ✓ | $0 \notin N$ ✗ | ✗ | ✗ |
| | $(N, \cdot)$ | ✓ | ✓ | $1 \in N$ ✓ | ✗ | ✗ |
| | $(N, -)$ | ✗ | ✗ | ✗ | ✗ | ✗ |
| | $(N, \div)$ | ✗ | ✗ | ✗ | ✗ | ✗ |
| $Z$ | $(Z, +)$ | ✓ | ✓ | $0 \in Z$, ✓ | ✓ | ✓ |
| | $(Z, \cdot)$ | ✓ | ✓ | $1 \in Z$, ✓ | ✗ | ✗ |
| | $(Z, -)$ | ✓ | ✗ | ✗ | ✗ | ✗ |
| | $(Z, \div)$ | ✗ | ✗ | ✗ | ✗ | ✗ |
| $Q$ | $(Q, +)$ | ✓ | ✓ | $0 \in Q$, ✓ | ✓ | ✓ |
| | $(Q, \cdot)$ | ✓ | ✓ | $1 \in Q$, ✓ | ✗, $0 \in Q$, | ✗ |
| | $(Q, -)$ | ✓ | ✗ | ✗ | ✗ | ✗ |
| | $(Q, \div)$ | ✗ | ✗ | ✗ | ✗ | ✗ |
| $Q^*$ | $(Q^*, +)$ | ✗ | ✗ | ✗ | ✗ | ✗ |
| | $(Q^*, \cdot)$ | ✓ | ✓ | $1 \in Q^*$, ✓ | ✓ | ✓ |
| | $(Q^*, -)$ | ✗ | ✗ | ✗ | ✗ | ✗ |
| | $(Q^*, \div)$ | ✓ | ✗ | ✗ | ✗ | ✗ |

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## Note

① Inverse of identity element will be identity element itself.

② Identity element (if exists) in the set, then it will be unique w.r.t. given binary opⁿ '$*$'.

③ In a group inverse of every element of the set exists within the set itself.

and if $inv(a) = b$ then $inv(b) = a$

$a^{-1} = b$ $\quad$ $b^{-1} = a$

$\boxed{\text{i.e. } (a^{-1})^{-1} = a}$

④ In a group $(G, *)$,

$(a*b)^{-1} = b^{-1} * a^{-1}$ always holds true $\forall a, b \in G$

## Groupoid

1. Closed

## Semi-group

1. Closed
2. Associative

## Monoid

1. Closed
2. Associative
3. Identity element must be present within the set

## Group

1. Closed
2. Associative
3. Identity element must be present within the set
4. Inverse of every element of the set must be present within the set

## Abelian group

1. Closed
2. Associative
3. Identity element must be present within the set
4. Inverse of every element of the set must be present within the set
5. Commutative

H.W.

#Q. Let $A = \{0, \pm 2, \pm 4, \pm 6, .....\}$

$B = \{0 \pm 1, \pm 3, \pm 5 .....\}$

Which of the following is not a semi- group

A (A, +)

B (A, •)

C (B, +)

D (B, •)

H.W.

#Q. Consider the set $\Sigma^*$ of all strings over the alphabet $\Sigma = \{0, 1\}$. $\Sigma^*$ with the concatenation operator for strings

A Not a semigroup

B Semi group but not a monoid

C Monoid but not a group.

D A group

H.W

#Q. Let A be the set of all non-singular matrices of order $n \times n$ over real number and let * be the matrix multiplication operation. Then

**A** A is closed under * but $\langle A, * \rangle$ is not a semigroup

**B** $\langle A, * \rangle$ is a semigroup but not a monoid.

**C** $\langle A, * \rangle$ is a monoid but not a group.

**D** $\langle A, * \rangle$ is a group but not an abelian group. nxdn n

H.W.

#Q. Let S be any finite set, and F(s) is defined as set of all function on set S. Then F(s) with respect to function composition operation (ie., o) is.

A. Not a semigroup

B. Semi group but not a monoid

C. Monoid but not a group.

D. A group

H.W.

#Q. Let Z is the set of all integers. The binary operation * is defined as a*b = max (a, b) then the structure (Z,*) is

A. Not a semigroup

B. Semi group but not a monoid

C. Monoid but not a group.

D. A group

H.W.

#Q. Let $Q^+$ be the set of all positive rational numbers. The binary operation * is defined as $\boxed{a * b = \frac{ab}{3} \forall a, b, \in Q^+}$ If $(Q^+, *)$ is a group then find

(i) identity element of the group

(ii) inverse of any element a, ∀, ∈ Group

H.W.

#Q. Which of the following statement is/are not true.

A $\{0, \pm 2k, \pm 4k,, \pm 6k, \ldots.\}$ is a group with respect to addition where k is any fixed positive integer

B $\{x \mid x \text{ is real number and } 0 < x \leq 1\}$ is a group with respect to multiplication

C $\{2^n \mid n \text{ is an integer}\}$ is a group with respect to multiplication

D None of these

**Note :-** Let 'S' is any non-Empty set.

Set 'S' is a group w.r.t. binary opⁿ '$*$' if and only if,

① $\boxed{a * b^{-1} \in S, \ \forall a, b \in S}$

and ② $\boxed{\text{'}*\text{' is associative}}$

This statement is sufficient to prove that,

① 'S' is closed w.r.t. '$*$'

② Identity w.r.t. '$*$' $\in S$

③ $a^{-1} \in S, \ \forall a \in S$

$\boxed{\text{Proof on next slide}}$

* Let $\boxed{a * b^{-1} \in S, \quad \forall a, b \in S.}$

① Identity :.

Let $a \in S$

$\therefore a, a \in S$

We know $\forall a, b \in S, a * b^{-1} \in S$

$\therefore$ if $a, a \in S$ then $\underbrace{a * a^{-1}}_{\downarrow} \in S$

$\boxed{e \in S}$

i.e., identity element will be present in set S.

② Inverse :.

* If identity element 'e' is the only element in the set, then we know $inv(e) = e$.

$\therefore$ Inverse exist for Every element of the set.

* If some other element $a \in S$, such that $a \neq e$

then $\underbrace{e, a \in S}$ {We already Proved that $e \in S$}

$\therefore \underbrace{e * (a^{-1})}_{\downarrow} \in S$

$\boxed{a^{-1} \in S}$

i.e, if $a \in S$ then $a^{-1} \in S$

③ Closed.

Let $a, b \in S$

We just proved that if $a, b \in S$, then $a^{-1}, b^{-1} \in S$

i.e, if $a, b \in S$ then $\underbrace{a, b^{-1} \in S}$ is also true

$\therefore a * \underbrace{(b^{-1})^{-1}}_{\downarrow} \in S$

$a * b \in S$

if $a, b \in S$ then $a * b \in S$

Hence Closed

## Finite Group

- A group $(G, *)$ is called a finite group if and only if set 'G' is a non-Empty finite set

- If $(G, *)$ is a finite group, then Order of group $(G, *)$ is same as Cardinality of set G!

i.e. $\boxed{O(G) = |G|}$

Order of group G

Note:

1. $\{0\}$ is a finite group of order $= 1$, w.r.t. addition
2. $\{1\}$ is a finite group of order $= 1$, w.r.t. multiplication.
3. $\{-1, 1\}$ is a finite group of order $= 2$, w.r.t. multiplication.

The above three groups are the only finite groups of real numbers w.r.t. addition and/or multiplication.

**Note:** If element 'e' is an identity element w.r.t. binary oph '$*$', then $\{e\}$ is always a finite group of 'order $= 1$' w.r.t. oph '$*$'.

**Note:** In a finite group of order $= 2$, Every element is inverse of itself.

eg: $1, \omega$ & $\omega^2$ are Cube roots of unity where $\boxed{\omega^3 = 1}$

The set $\{1, \omega, \omega^2\}$ form a finite group of order $= 3$ w.r.t. binary opn multiplication.

To find inverse we can use the concept of Composition table

① We know $\omega^3 = 1$,
$\therefore$ Set is closed w.r.t. multiplication

② Multiplication is associative

③ Identity w.r.t. multiplication is '1' and $1 \in$ Set.

④ Inverse, $inv(1) = 1$, $inv(\omega) = \omega^2$, $inv(\omega^2) = \omega$ } inverse exist for Every element

opn is multiplication

| $\cdot$ | $1$ | $\omega$ | $\omega^2$ |
|---|---|---|---|
| $1$ | $1 = e$ | $\omega$ | $\omega^2$ |
| $\omega$ | $\omega$ | $\omega^2$ | $\omega^3 = 1 = e$ |
| $\omega^2$ | $\omega^2$ | $\omega^3 = 1 = e$ | |

$inv(1) = 1$

$inv(\omega) = \omega^2$

$inv(\omega^2) = \omega$

$\omega^4 = \omega^3 \omega = 1 \cdot \omega = \omega$

eg:- $1, -1, j$ & $-j$ are four roots of Unity.

The set $\{1, -1, j, -j\}$ form a finite group of order = 4, w.r.t. binary opⁿ multiplication.

1. $j^2 = -1$, ∴ Closed
2. Multiplication is associative
3. Identity, i.e. $1 \in$ Set
4. Inverse,

$inv(1) = 1$

$inv(-1) = -1$

$inv(j) = -j$

$inv(-j) = j$

inverse exist for every element.

| multiplication | $1$ | $-1$ | $j$ | $-j$ |
|---|---|---|---|---|
| $1$ | $1$ | $-1$ | $j$ | $-j$ |
| $-1$ | $-1$ | $1 = e$ | $-j$ | $+j$ |
| $j$ | $j$ | $-j$ | $j^2 = -1$ | $-j^2 = +1 = e$ |
| $-j$ | $-j$ | $+j$ | $-j^2 = 1 = e$ | $+j^2 = -1$ |

* **Note**:- The set of 'n' roots of Unity will form a finite group of order $= n$ w.r.t. multiplication.

let $\phi$ is . root of 'n' roots of unity.

$$\underbrace{1, \phi_1, \phi_2 \ldots \phi_n}_{\text{Multiplication of roots}} = (-1)^{n+1}$$

$$\{1, \omega, \omega^2\}$$

$$1 \cdot \omega \cdot \omega^2 = (-1)^{3+1}$$

$$\boxed{\omega^3 = 1}$$

$$\underline{1, -1, j, -j.}$$

$$(1) \cdot (-1) \cdot (j) \cdot (-j) = (-1)^{4+1}$$

$$j^2 = -1$$

# Topic : Addition modulo 'm' $\oplus_m$

it gives remainder

- Let 'm' is a fixed positive integer.
- For any two non-negative integers $a$ & $b$

$$a \oplus_m b = \begin{cases} a+b & \text{if } (a+b) < m \\ r & \text{otherwise} \end{cases}$$

$|||$

$(a+b) \% m$

$r$ is the remainder obtained on dividing $(a+b)$ by $m$

e.g.

$$\boxed{\begin{array}{l} 2 \oplus_6 3 = 5 \\ 4 \oplus_6 5 = 3 \end{array}}$$

# Topic : Multiplication modulo 'm' $\otimes_m$

it gives remainder

- Let $m$ is a fixed positive integer
- For any two non-negative integer $a$ & $b$

$$a \otimes_m b = \begin{cases} (a \cdot b) & \text{if } (a \cdot b) < m \\ r & \text{Otherwise} \end{cases}$$

(remainder)

$$\equiv (a \cdot b) \% m$$

e.g. $\boxed{\begin{array}{l} 2 \otimes_7 3 = 6 \\ 4 \otimes_7 5 = 6 \end{array}}$

Topic : NOTE

(1) The set $\{0, 1, 2, \ldots\ldots, (n-1)\}$ form a finite group of order $= n$ w.r.t. binary opⁿ $\oplus_n$

eg: $\{0, 1, 2, 3, 4\}$ is a finite group of order $= 5$ w.r.t $\oplus_5$

(1) Closed ✓

(2) $\oplus_n$ is associative for any positive integer 'n'

(3) Identity w.r.t. $\oplus_n = 0 \in$ Set ✓

(4) inverse:- $inv(0) = 0$, $inv(1) = 4$, $inv(2) = 3$, $inv(3) = 2$, $inv(4) = 1$

$inv(0) = 0$

$inv(x) = (n - x) \quad \forall x \neq 0$

**Note:-** let 'n' is given

i.e, $x$ & $n$ are Co-prime
or "relatively prime"

let $A = \{x \mid x \in N \text{ and } x < n \text{ and } \underbrace{GCD(x,n) = 1}\}$

$N = \{1, 2, 3, \ldots\}$

Here, A is a set of all natural numbers that are less than 'n' and Co-prime to 'n'

eg. let $n = 15$, then $A = ?$

$$A = \{1, 2, 4, 7, 8, 11, 13, 14\}$$

Topic : NOTE

(2) The set of all natural numbers that are less than 'n' and Co-prime to 'n' definitely from a group w.r.t. multiplication modulo 'n' {i.e; $\otimes_n$}

eg. let $n = 15$, then

$\{1, 2, 4, 7, 8, 11, 13, 14\}$ is group of order $= 8$ w.r.t. binary opn $\otimes_{15}$

① Closed ✓

② $\otimes_n$ is associative for any +ve integer 'n'

③ Identity w.r.t $\otimes_n = 1$

④ Inverse:-

inv(1) = 1 $\quad 1 \otimes_{15} 1 = 1$

inv(2) = 8 $\quad 2 \otimes_{15} 8 = 1$

inv(4) = 4 $\quad 4 \otimes_{15} 4 = 1$

inv(7) = 13 $\quad 7 \otimes_{15} 13 = 1$

inv(8) = 2

inv(11) = 11 $\quad 11 \otimes_{15} 11 = 1$

inv(13) = 7

inv(14) = 14 $\quad 14 \otimes_{15} 14 = 1$

eg. Let $n = 8$, then

$\{1, 3, 5, 7\}$ is a finite group of order $= 4$ w.r.t. $\otimes_8$

① Closed ✓

② Associative ✓

③ Identity $\in S$

④ Inverse

$inv(1) = 1$ $\quad 1 \otimes_8 1 = 1$

$inv(3) = 3$ $\quad 3 \otimes_8 3 = 1$

$inv(5) = 5$ $\quad 5 \otimes_8 5 = 1$

$inv(7) = 7$ $\quad 7 \otimes_8 7 = 1$

In this group every element is inverse of itself eg: ① $\{1, 3, 5, 7\}$ is a group of order $= 4$ w.r.t. $\otimes_8$

$\{1\}$ is a finite group of order $= 1$ w.r.t. $\otimes_8$

Note:

- $\{1, 3\}$ is a finite group of order $= 2$ w.r.t. $\otimes_8$
- $\{1, 5\}$ is a finite group of order $= 2$ w.r.t. $\otimes_8$
- $\{1, 7\}$ is a finite group of order $= 2$ w.r.t. $\otimes_8$

← some subsets of original set may also be the group w.r.t. same binary opⁿ

But all subsets need not be the group w.r.t. same binary opⁿ.

eg. $\{3, 5\}$ is not closed w.r.t. $\otimes_8$ $\therefore$ it is not a group.

eg. $\{1, 3, 5\}$ is not closed w.r.t. $\otimes_8$ $\therefore$ it is not a group.

eg. We know $\{0, 1, 2, 3\}$ is a group w.r.t. $\oplus_4$

Here

- $\{0\}$ is also a group w.r.t. $\oplus_4$ ✓
- $\{0, 2\}$ is also a group w.r.t. $\oplus_4$
- $\{0, 1\}$ is not closed w.r.t. $\oplus_4$ $\therefore$ not a group

**Note:-** We know,

The set of all natural numbers that are less than n and co-prime to n form a group w.r.t. binary opⁿ $\otimes_n$

∴ if 'p' is a prime number then the set $\{1, 2, 3, \ldots, (p-1)\}$ will form a finite group of order $= (p-1)$ w.r.t. binary opⁿ $\otimes_p$

eg: The set $\{1, 2, 3, 4, 5, 6\}$ is a group of order $= 6$ w.r.t. opⁿ $\otimes_7$

H.W.

Q: Which of the following is/are not true?

(a) $\{1, 2, 3, 4, 5\}$ is a group w.r.t. $\oplus_6$

(b) $\{1, 2, 3, 4, 5$ is a group w.r.t. $\otimes_6$

(c) $\{0, 1, 2, 3, 4\}$ is a group w.r.t. $\oplus_5$

(d) $\{0, 1, 2, 3, 4\}$ is a group w.r.t. $\otimes_5$

# 2 mins Summary

- **Topic** Basic concepts associated with group
- **Topic** Finite groups
- **Topic** Addition modulo m
- **Topic** Multiplication modulo m

Slide

# THANK - YOU

Slide